In [1]:
```
import mysql.connector
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

# Set style for all charts
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("Set2")
plt.rcParams['figure.figsize'] = (12, 6)
plt.rcParams['font.size'] = 12

print("✅ Libraries loaded successfully")
```

```
Matplotlib is building the font cache; this may take a moment.
✅ Libraries loaded successfully
```

In [3]:
```
conn = mysql.connector.connect(
    host='127.0.0.1',
    user='root',
    password='Alenwalker10@',   # ← Replace with your MySQL passwor
    database='adarsha_clinic'
)
cursor = conn.cursor()
print("✅ Connected to adarsha_clinic database")
```

```
✅ Connected to adarsha_clinic database
```

In [4]:
```
query = """
SELECT 'patients' AS table_name, COUNT(*) AS row_count FROM patient
UNION ALL
SELECT 'visits', COUNT(*) FROM visits
UNION ALL
SELECT 'billing', COUNT(*) FROM billing
UNION ALL
SELECT 'panchakarma_sessions', COUNT(*) FROM panchakarma_sessions
UNION ALL
SELECT 'therapy_pricing', COUNT(*) FROM therapy_pricing
"""
df = pd.read_sql(query, conn)
print("📊 Adarsha Clinic — Database Summary")
print("=" * 40)
print(df.to_string(index=False))
print(f"\n📈 Total rows: {df['row_count'].sum():,}")
```

```
📊 Adarsha Clinic — Database Summary
========================================
          table_name  row_count
            patients       3000
              visits       8977
             billing       8977
panchakarma_sessions       8789
     therapy_pricing          9

📈 Total rows: 29,752
```

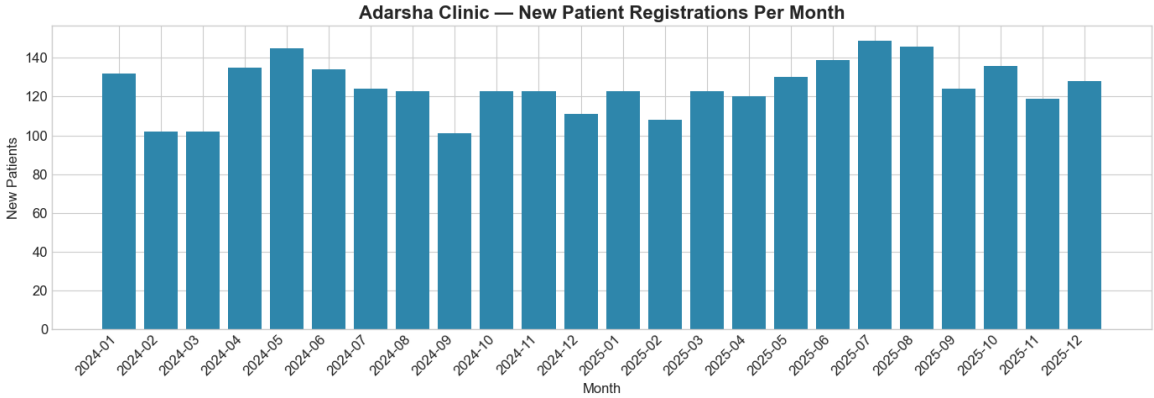
In [5]:

```
patients = pd.read_sql("SELECT * FROM patients", conn)
visits = pd.read_sql("SELECT * FROM visits", conn)
billing = pd.read_sql("SELECT * FROM billing", conn)
panchakarma = pd.read_sql("SELECT * FROM panchakarma_sessions", con
therapy_pricing = pd.read_sql("SELECT * FROM therapy_pricing", conn

print(f"✅ patients: {len(patients)} rows")
print(f"✅ visits: {len(visits)} rows")
print(f"✅ billing: {len(billing)} rows")
print(f"✅ panchakarma: {len(panchakarma)} rows")
print(f"✅ therapy_pricing: {len(therapy_pricing)} rows")
```

```
✅ patients: 3000 rows
✅ visits: 8977 rows
✅ billing: 8977 rows
✅ panchakarma: 8789 rows
✅ therapy_pricing: 9 rows
```

In [6]:

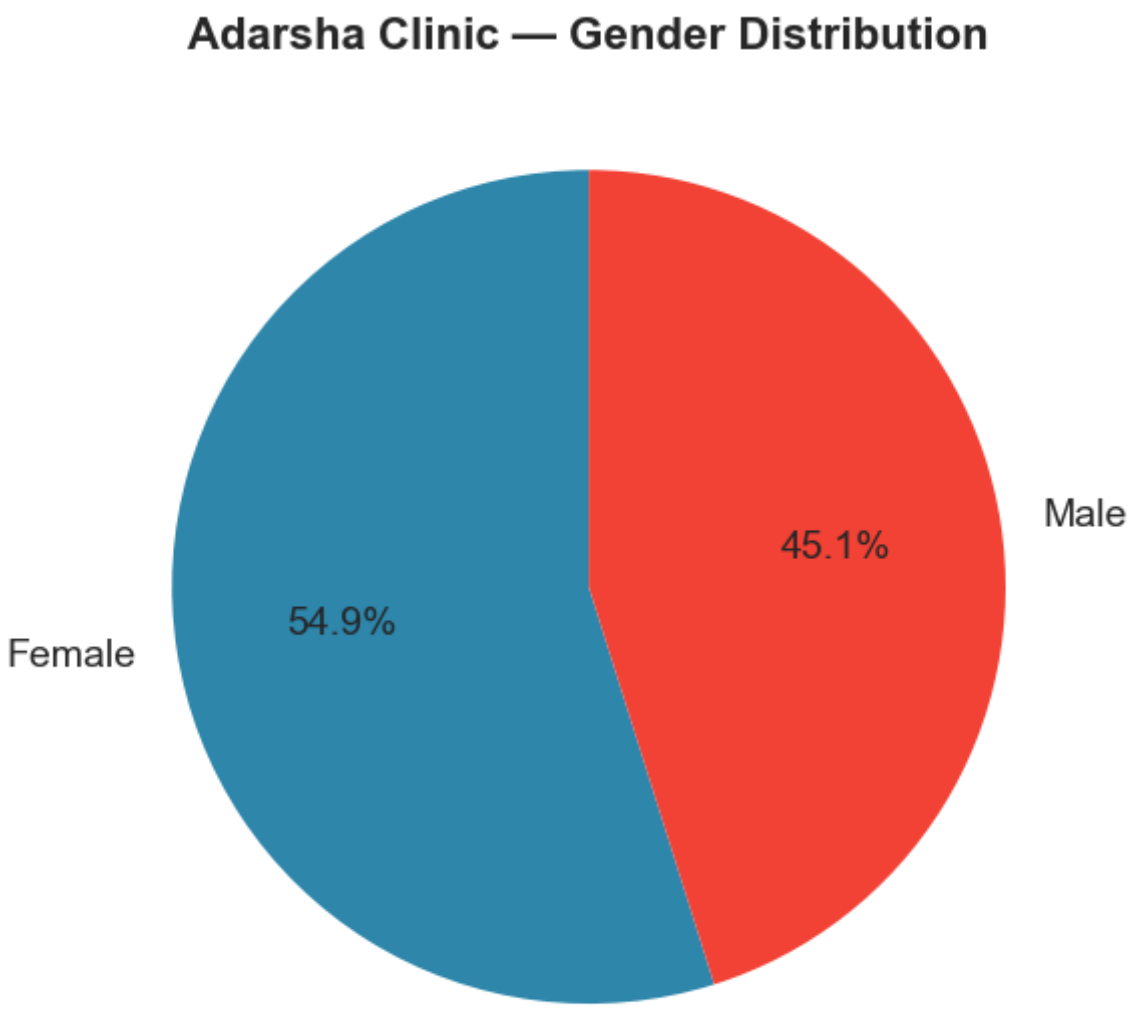
```
query = """
SELECT
    DATE_FORMAT(registration_date, '%Y-%m') AS month,
    COUNT(*) AS new_patients
FROM patients
GROUP BY month
ORDER BY month
"""
df1 = pd.read_sql(query, conn)

plt.figure(figsize=(14, 5))
plt.bar(df1['month'], df1['new_patients'], color='#2E86AB')
plt.title('Adarsha Clinic — New Patient Registrations Per Month', f
plt.xlabel('Month')
plt.ylabel('New Patients')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print(df1.to_string(index=False))
```




```
  month  new_patients
2024-01           132
2024-02           102
2024-03           102
2024-04           135
2024-05           145
2024-06           134
2024-07           124
2024-08           123
2024-09           101
2024-10           123
2024-11           123
2024-12           111
2025-01           123
2025-02           108
2025-03           123
2025-04           120
2025-05           130
2025-06           139
2025-07           149
2025-08           146
2025-09           124
2025-10           136
2025-11           119
2025-12           128
```

In [7]:

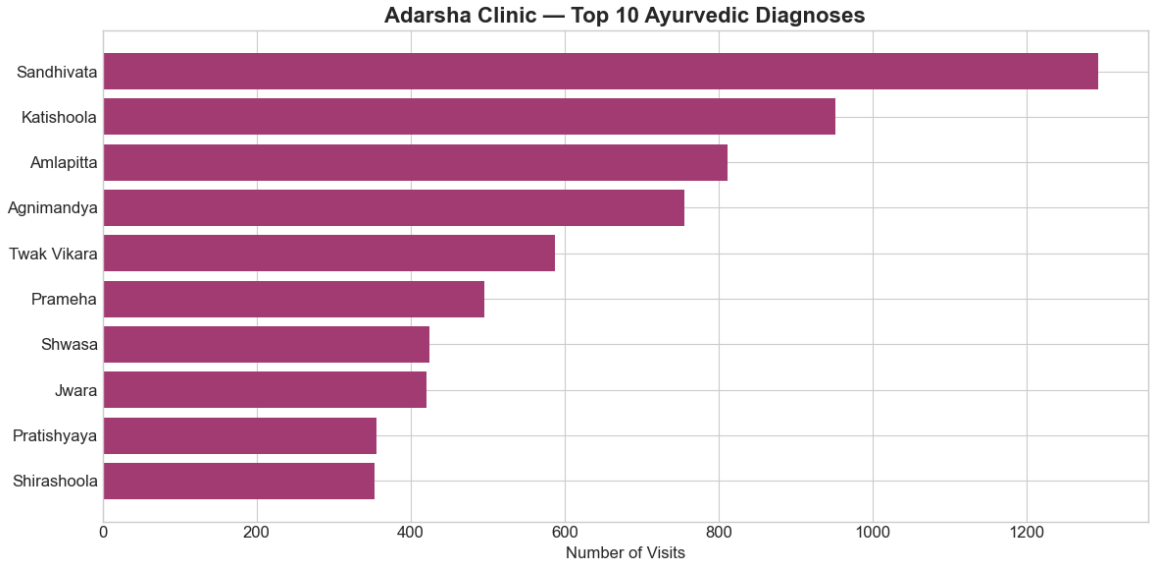
```
query = """
SELECT
    gender,
    COUNT(*) AS count,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM patients), 1) AS
FROM patients
GROUP BY gender
"""
df2 = pd.read_sql(query, conn)

fig, ax = plt.subplots(figsize=(6, 6))
colors = ['#2E86AB', '#F24236']
ax.pie(df2['count'], labels=df2['gender'], autopct='%1.1f%%', color
ax.set_title('Adarsha Clinic — Gender Distribution', fontsize=16, f
plt.tight_layout()
plt.show()

print(df2.to_string(index=False))
```




```
gender  count  percentage
Female   1646        54.9
  Male   1354        45.1
```

In [9]:

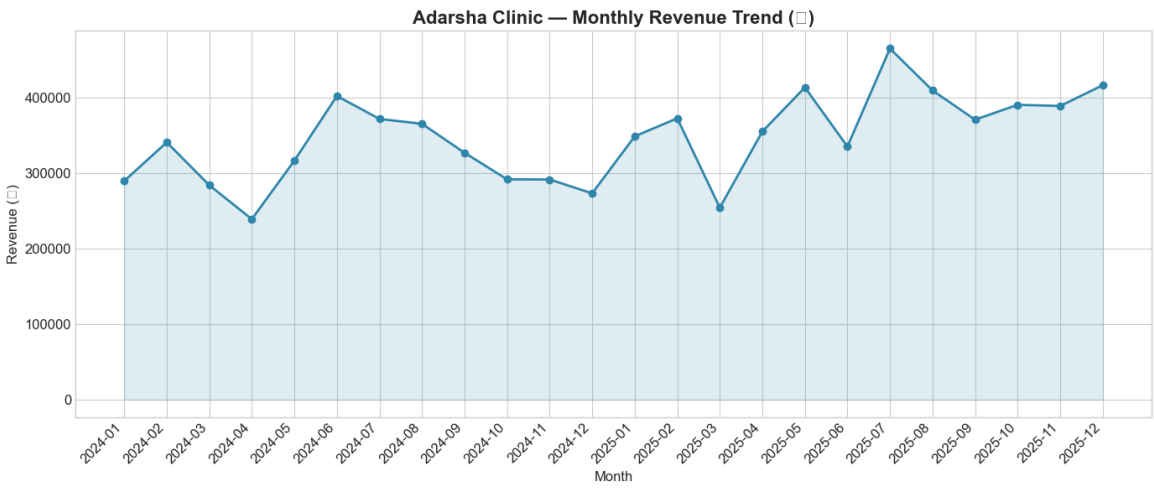
```
query = """
SELECT
    ayurvedic_diagnosis,
    COUNT(*) AS visit_count
FROM visits
GROUP BY ayurvedic_diagnosis
ORDER BY visit_count DESC
LIMIT 10
"""
df3 = pd.read_sql(query, conn)

plt.figure(figsize=(12, 6))
plt.barh(df3['ayurvedic_diagnosis'][::-1], df3['visit_count'][::-1]
plt.title('Adarsha Clinic — Top 10 Ayurvedic Diagnoses', fontsize=1
plt.xlabel('Number of Visits')
plt.tight_layout()
plt.show()

print(df3.to_string(index=False))
```




```
ayurvedic_diagnosis  visit_count
         Sandhivata         1293
         Katishoola          951
          Amlapitta          812
         Agnimandya          755
        Twak Vikara          587
            Prameha          496
             Shwasa          424
              Jwara          420
        Pratishyaya          356
        Shirashoola          353
```

In [10]:

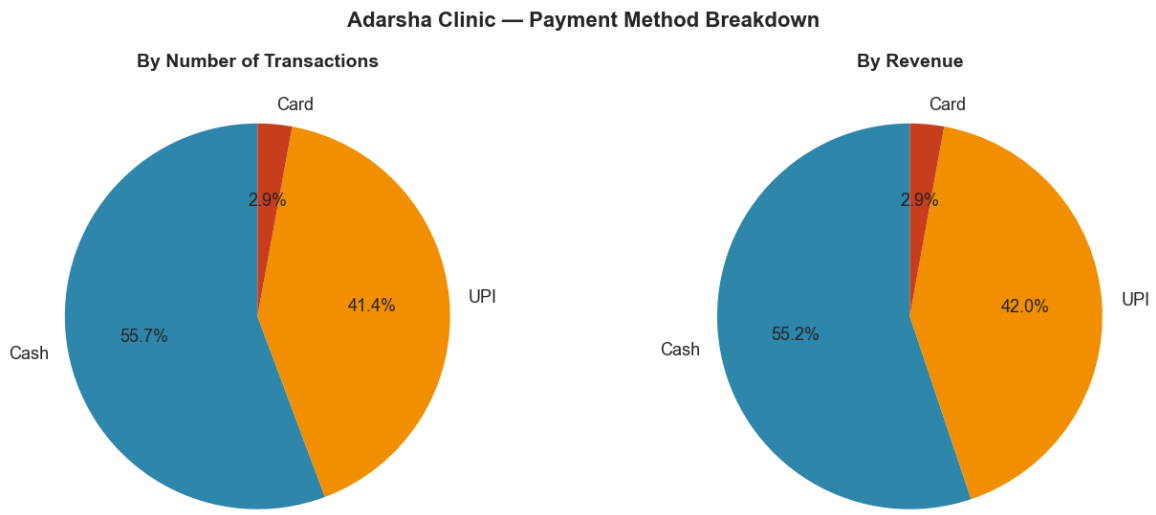
```
query = """
SELECT
    DATE_FORMAT(bill_date, '%Y-%m') AS month,
    SUM(total_amount) AS revenue,
    COUNT(*) AS total_bills,
    ROUND(AVG(total_amount), 2) AS avg_bill
FROM billing
GROUP BY month
ORDER BY month
"""
df4 = pd.read_sql(query, conn)

plt.figure(figsize=(14, 6))
plt.plot(df4['month'], df4['revenue'], marker='o', linewidth=2, col
plt.fill_between(range(len(df4)), df4['revenue'], alpha=0.15, color
plt.title('Adarsha Clinic — Monthly Revenue Trend (₹)', fontsize=16
plt.xlabel('Month')
plt.ylabel('Revenue (₹)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

total_revenue = df4['revenue'].sum()
print(f"💰 Total Revenue (24 months): ₹{total_revenue:,.0f}")
print(f"📊 Average Monthly Revenue: ₹{df4['revenue'].mean():,.0f}")
```




```
💰 Total Revenue (24 months): ₹8,311,647
📊 Average Monthly Revenue: ₹346,319
```

In [11]:

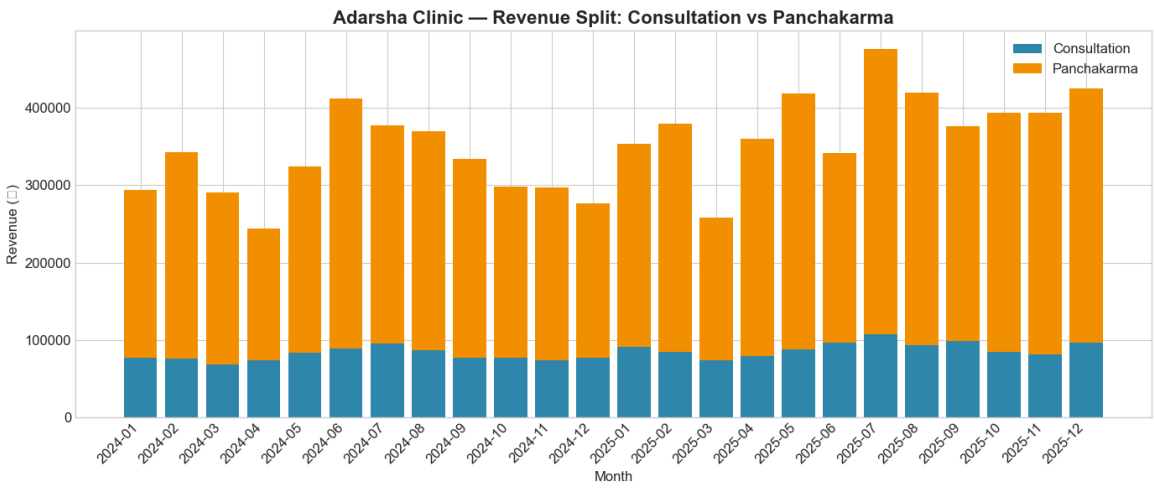
```
query = """
SELECT
    payment_method,
    COUNT(*) AS count,
    SUM(total_amount) AS total_revenue,
    ROUND(AVG(total_amount), 2) AS avg_bill
FROM billing
GROUP BY payment_method
ORDER BY total_revenue DESC
"""
df5 = pd.read_sql(query, conn)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))
colors = ['#2E86AB', '#F18F01', '#C73E1D']

ax1.pie(df5['count'], labels=df5['payment_method'], autopct='%1.1f%
ax1.set_title('By Number of Transactions', fontsize=14, fontweight=

ax2.pie(df5['total_revenue'], labels=df5['payment_method'], autopct
ax2.set_title('By Revenue', fontsize=14, fontweight='bold')

plt.suptitle('Adarsha Clinic — Payment Method Breakdown', fontsize=
plt.tight_layout()
plt.show()

print(df5.to_string(index=False))
```



payment_method	count	total_revenue	avg_bill
Cash	5000	4584714.0	916.94
UPI	3718	3489677.0	938.59
Card	259	237256.0	916.05

In [12]:

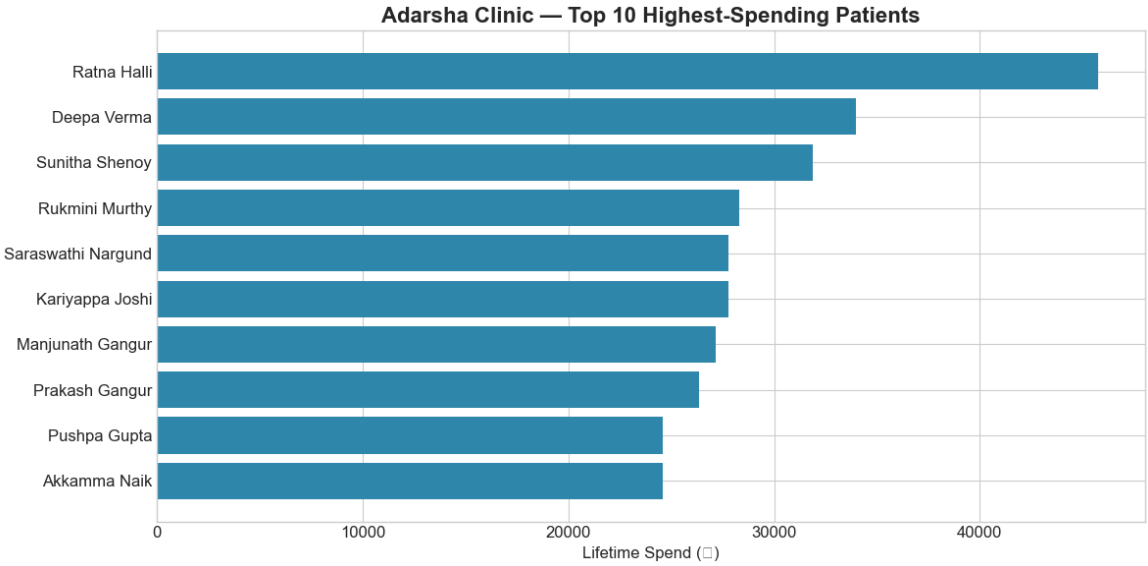
```
query = """
SELECT
    DATE_FORMAT(bill_date, '%Y-%m') AS month,
    SUM(consultation_fee) AS consultation_revenue,
    SUM(therapy_charges) AS panchakarma_revenue,
    SUM(total_amount) AS total_revenue
FROM billing
GROUP BY month
ORDER BY month
"""
df6 = pd.read_sql(query, conn)

plt.figure(figsize=(14, 6))
x = range(len(df6))
plt.bar(x, df6['consultation_revenue'], label='Consultation', color
plt.bar(x, df6['panchakarma_revenue'], bottom=df6['consultation_rev
plt.title('Adarsha Clinic — Revenue Split: Consultation vs Panchaka
plt.xlabel('Month')
plt.ylabel('Revenue (₹)')
plt.xticks(x, df6['month'], rotation=45, ha='right')
plt.legend()
plt.tight_layout()
plt.show()

total_consult = df6['consultation_revenue'].sum()
total_therapy = df6['panchakarma_revenue'].sum()
print(f"Consultation Revenue: ₹{total_consult:,.0f} ({total_consult
print(f"Panchakarma Revenue: ₹{total_therapy:,.0f} ({total_therapy*
```




```
Consultation Revenue: ₹2,026,550 (24.0%)
Panchakarma Revenue: ₹6,433,150 (76.0%)
```

In [13]:

```
query = """
SELECT
    p.patient_id,
    CONCAT(p.first_name, ' ', p.last_name) AS patient_name,
    p.village_area,
    COUNT(v.visit_id) AS total_visits,
    SUM(b.total_amount) AS lifetime_spend
FROM patients p
JOIN visits v ON p.patient_id = v.patient_id
JOIN billing b ON v.visit_id = b.visit_id
GROUP BY p.patient_id, patient_name, p.village_area
ORDER BY lifetime_spend DESC
LIMIT 10
"""
df7 = pd.read_sql(query, conn)

plt.figure(figsize=(12, 6))
plt.barh(df7['patient_name'][::-1], df7['lifetime_spend'][::-1], co
plt.title('Adarsha Clinic — Top 10 Highest-Spending Patients', font
plt.xlabel('Lifetime Spend (₹)')
plt.tight_layout()
plt.show()

print(df7.to_string(index=False))
```

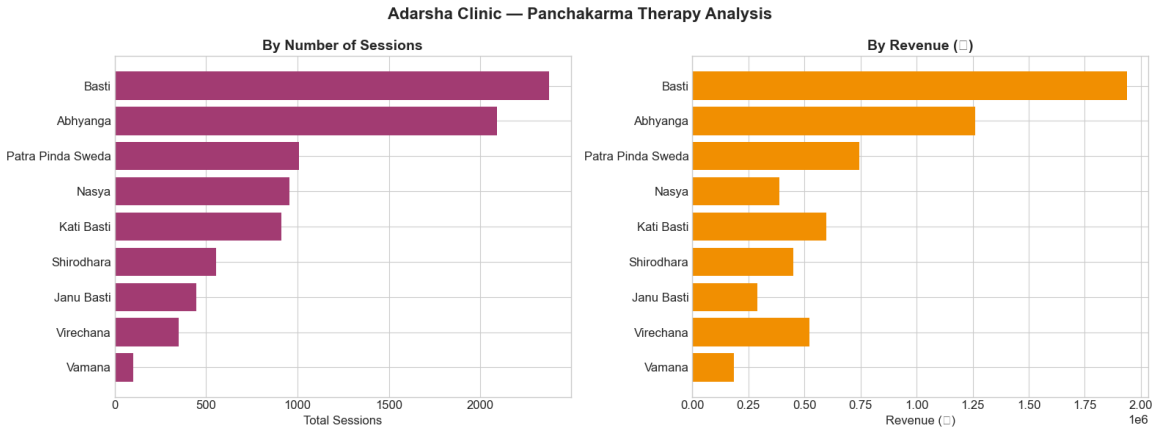



```
 patient_id       patient_name        village_area  total_visits  li
fetime_spend
       1049        Ratna Halli  Harapanahalli Town             9
45750.0
       3046        Deepa Verma Hagari Bommanahalli            12
33992.0
       3918     Sunitha Shenoy              Haveri             7
31903.0
       1972     Rukmini Murthy            Mundargi            15
28300.0
       3209 Saraswathi Nargund        Hire Benakal             9
27800.0
       1785    Kariyappa Joshi              Koppal            12
27800.0
       2144   Manjunath Gangur  Harapanahalli Town             7
27187.0
       3978     Prakash Gangur         Tekkalakote            12
26335.0
       3307       Pushpa Gupta              Koppal            12
24600.0
       1799       Akkamma Naik              Sandur             7
24585.0
```

In [14]:

```
query = """
SELECT
    therapy_name,
    COUNT(*) AS total_sessions,
    COUNT(DISTINCT patient_id) AS unique_patients,
    SUM(session_fee) AS total_revenue
FROM panchakarma_sessions
GROUP BY therapy_name
ORDER BY total_sessions DESC
"""
df8 = pd.read_sql(query, conn)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))

ax1.barh(df8['therapy_name'][::-1], df8['total_sessions'][::-1], co
ax1.set_title('By Number of Sessions', fontsize=14, fontweight='bol
ax1.set_xlabel('Total Sessions')
```

```
ax2.barh(df8['therapy_name'][::-1], df8['total_revenue'][::-1], col
ax2.set_title('By Revenue (₹)', fontsize=14, fontweight='bold')
ax2.set_xlabel('Revenue (₹)')

plt.suptitle('Adarsha Clinic — Panchakarma Therapy Analysis', fonts
plt.tight_layout()
plt.show()

print(df8.to_string(index=False))
```

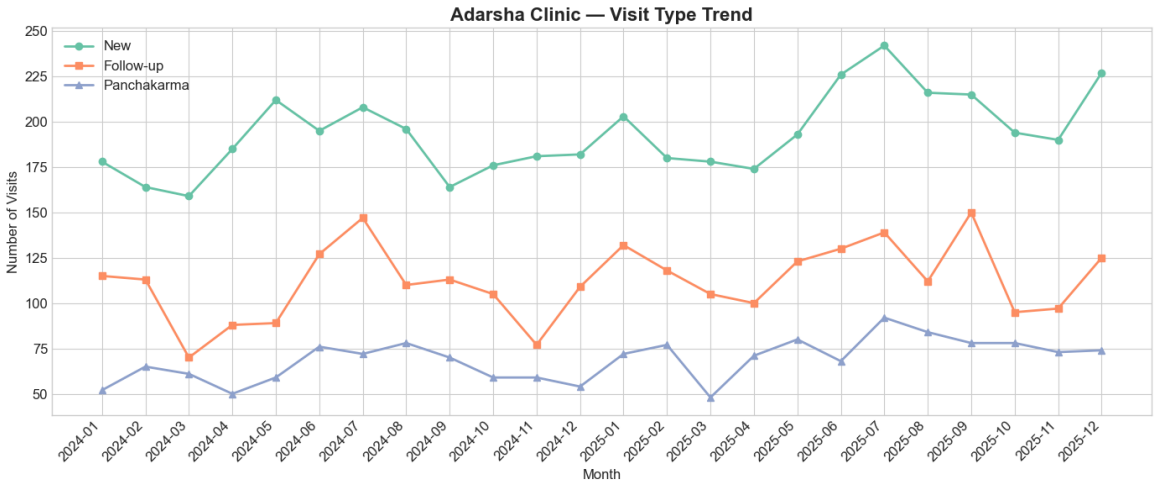



```
     therapy_name  total_sessions  unique_patients  total_revenue
            Basti            2375              301     1936746.36
        Abhyanga            2090              275     1262400.00
Patra Pinda Sweda           1008              143      745618.71
            Nasya            955              122      386999.31
      Kati Basti             912              133      596569.38
      Shirodhara             553               83      450856.36
      Janu Basti             445               64      291598.92
       Virechana             348              318      522000.00
          Vamana             103               98      185400.00
```

In [15]:

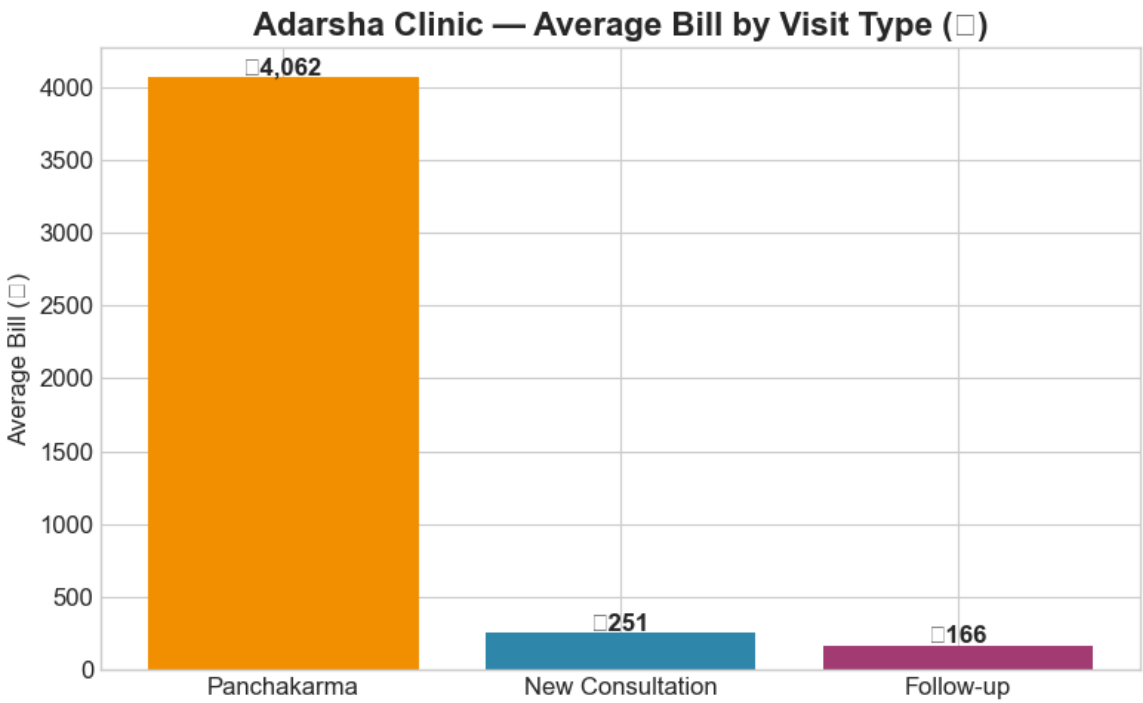
```
query = """
SELECT
    DATE_FORMAT(v.visit_date, '%Y-%m') AS month,
    SUM(CASE WHEN v.visit_type = 'New Consultation' THEN 1 ELSE 0 E
    SUM(CASE WHEN v.visit_type = 'Follow-up' THEN 1 ELSE 0 END) AS
    SUM(CASE WHEN v.visit_type = 'Panchakarma' THEN 1 ELSE 0 END) A
FROM visits v
GROUP BY month
ORDER BY month
"""
df9 = pd.read_sql(query, conn)

plt.figure(figsize=(14, 6))
x = range(len(df9))
plt.plot(x, df9['new_patients'], marker='o', label='New', linewidth
plt.plot(x, df9['follow_ups'], marker='s', label='Follow-up', linew
plt.plot(x, df9['panchakarma_visits'], marker='^', label='Panchakar
plt.title('Adarsha Clinic — Visit Type Trend', fontsize=16, fontwei
plt.xlabel('Month')
plt.ylabel('Number of Visits')
plt.xticks(x, df9['month'], rotation=45, ha='right')
plt.legend()
```

```
plt.tight_layout()
plt.show()
```



In [16]:

```
query = """
SELECT
    v.visit_type,
    COUNT(*) AS visits,
    ROUND(AVG(b.total_amount), 2) AS avg_bill,
    ROUND(MIN(b.total_amount), 2) AS min_bill,
    ROUND(MAX(b.total_amount), 2) AS max_bill
FROM visits v
JOIN billing b ON v.visit_id = b.visit_id
GROUP BY v.visit_type
ORDER BY avg_bill DESC
"""
df10 = pd.read_sql(query, conn)

plt.figure(figsize=(8, 5))
colors = ['#F18F01', '#2E86AB', '#A23B72']
plt.bar(df10['visit_type'], df10['avg_bill'], color=colors)
plt.title('Adarsha Clinic — Average Bill by Visit Type (₹)', fontsi
plt.ylabel('Average Bill (₹)')
for i, v in enumerate(df10['avg_bill']):
    plt.text(i, v + 20, f'₹{v:,.0f}', ha='center', fontweight='bold
plt.tight_layout()
plt.show()

print(df10.to_string(index=False))
```




```
      visit_type  visits  avg_bill  min_bill  max_bill
     Panchakarma    1650   4062.03     700.0   10800.0
New Consultation    4638    250.61     100.0     300.0
       Follow-up    2689    166.21      50.0     200.0
```

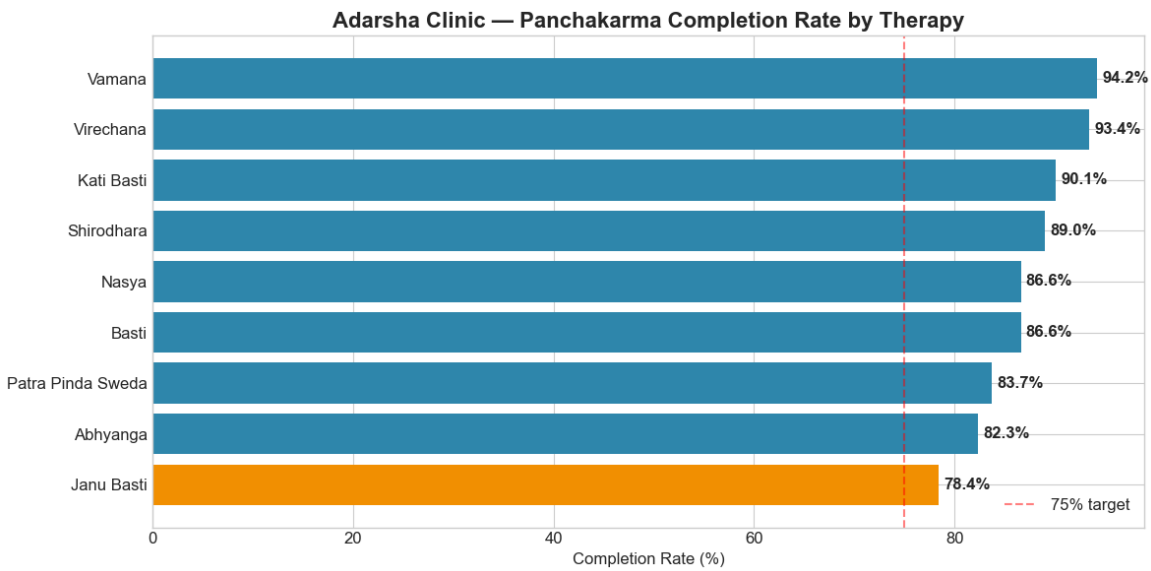
In [17]:

```
query = """
WITH monthly_rev AS (
    SELECT
        DATE_FORMAT(bill_date, '%Y-%m') AS month,
        SUM(total_amount) AS revenue
    FROM billing
    GROUP BY month
)
SELECT
    month,
    revenue,
    LAG(revenue) OVER (ORDER BY month) AS prev_month_revenue,
    ROUND((revenue - LAG(revenue) OVER (ORDER BY month)) * 100.0 /
        LAG(revenue) OVER (ORDER BY month), 1) AS growth_pct
FROM monthly_rev
ORDER BY month
"""
df11 = pd.read_sql(query, conn)

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 10))

ax1.plot(df11['month'], df11['revenue'], marker='o', linewidth=2, c
ax1.set_title('Monthly Revenue', fontsize=14, fontweight='bold')
ax1.set_ylabel('Revenue (₹)')
ax1.tick_params(axis='x', rotation=45)

colors = ['#2E86AB' if x >= 0 else '#C73E1D' for x in df11['growth_
ax2.bar(df11['month'], df11['growth_pct'].fillna(0), color=colors)
ax2.set_title('Month-over-Month Growth %', fontsize=14, fontweight=
ax2.set_ylabel('Growth %')
ax2.axhline(y=0, color='black', linewidth=0.5)
```

```
ax2.tick_params(axis='x', rotation=45)

plt.suptitle('Adarsha Clinic — Revenue Growth Analysis', fontsize=1
plt.tight_layout()
plt.show()
```

In [18]:

```
query = """
SELECT
    therapy_name,
    COUNT(DISTINCT CONCAT(patient_id, '-', visit_id)) AS total_pack
    SUM(CASE WHEN status = 'Completed' THEN 1 ELSE 0 END) AS comple
    COUNT(*) AS total_sessions,
    ROUND(SUM(CASE WHEN status = 'Completed' THEN 1 ELSE 0 END) * 1
FROM panchakarma_sessions
GROUP BY therapy_name
ORDER BY completion_rate ASC
"""
df12 = pd.read_sql(query, conn)

plt.figure(figsize=(12, 6))
colors = ['#C73E1D' if x < 70 else '#F18F01' if x < 80 else '#2E86A
plt.barh(df12['therapy_name'], df12['completion_rate'], color=color
plt.title('Adarsha Clinic — Panchakarma Completion Rate by Therapy'
plt.xlabel('Completion Rate (%)')
plt.axvline(x=75, color='red', linestyle='--', alpha=0.5, label='75
for i, v in enumerate(df12['completion_rate']):
    plt.text(v + 0.5, i, f'{v}%', va='center', fontweight='bold')
plt.legend()
plt.tight_layout()
plt.show()

print(df12.to_string(index=False))
```




```
     therapy_name  total_packages  completed_sessions  total_session
s  completion_rate
       Janu Basti              65               349.0             44
5            78.4
         Abhyanga             306              1721.0            209
0            82.3
Patra Pinda Sweda             147               844.0            100
8            83.7
           Basti             332              2057.0            237
5            86.6
           Nasya             125               827.0             95
5            86.6
       Shirodhara              85               492.0             55
3            89.0
       Kati Basti             139               822.0             91
2            90.1
        Virechana             348               325.0             34
8            93.4
          Vamana             103                97.0             10
3            94.2
```

In [19]:

```
query = """
WITH first_visit AS (
    SELECT patient_id, MIN(visit_date) AS first_date
    FROM visits
    GROUP BY patient_id
),
second_visit AS (
    SELECT v.patient_id, MIN(v.visit_date) AS return_date
    FROM visits v
    JOIN first_visit fv ON v.patient_id = fv.patient_id
    WHERE v.visit_date > fv.first_date
    AND v.visit_date <= DATE_ADD(fv.first_date, INTERVAL 90 DAY)
    GROUP BY v.patient_id
)
SELECT
    COUNT(DISTINCT fv.patient_id) AS total_first_visitors,
    COUNT(DISTINCT sv.patient_id) AS returned_within_90_days,
    ROUND(COUNT(DISTINCT sv.patient_id) * 100.0 / COUNT(DISTINCT fv
FROM first_visit fv
```

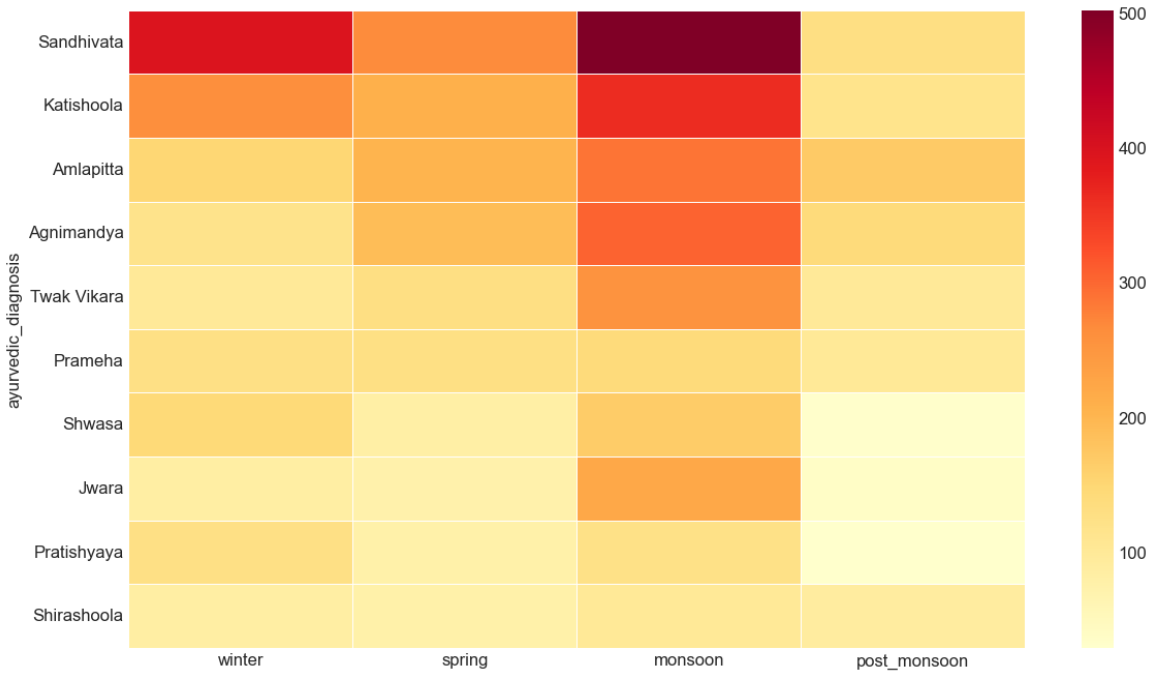
```
LEFT JOIN second_visit sv ON fv.patient_id = sv.patient_id
"""
df13 = pd.read_sql(query, conn)

print("📊 Adarsha Clinic — Patient Retention Analysis")
print("=" * 50)
print(f"Total first-time visitors: {df13['total_first_visitors'].va
print(f"Returned within 90 days:   {df13['returned_within_90_days']
print(f"Retention rate:            {df13['retention_rate_pct'].valu
```

```
📊 Adarsha Clinic — Patient Retention Analysis
==================================================
Total first-time visitors: 3,000
Returned within 90 days:   1,284
Retention rate:            42.8%
```

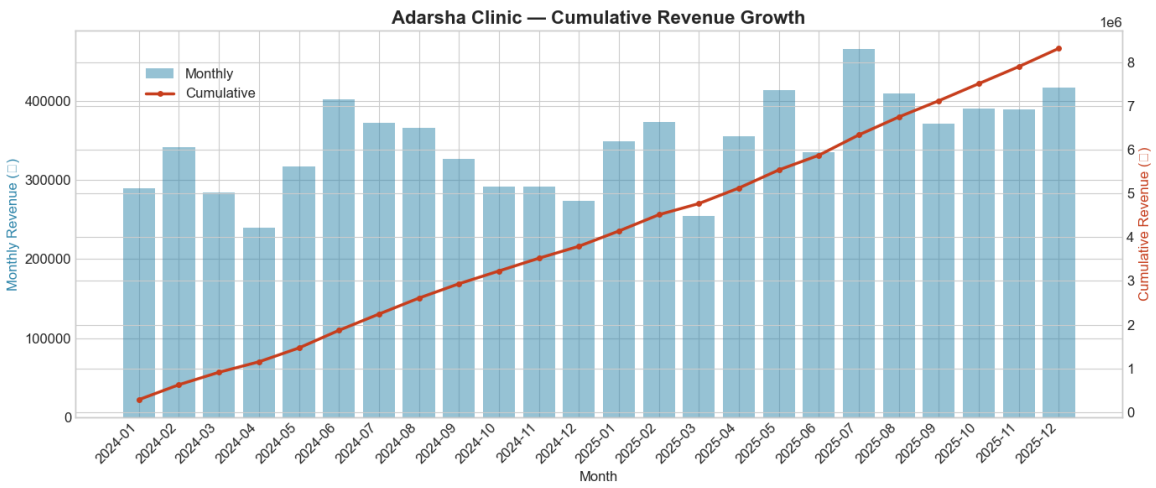
In [20]:

```
query = """
SELECT
    ayurvedic_diagnosis,
    SUM(CASE WHEN MONTH(visit_date) IN (12,1,2) THEN 1 ELSE 0 END)
    SUM(CASE WHEN MONTH(visit_date) IN (3,4,5) THEN 1 ELSE 0 END) A
    SUM(CASE WHEN MONTH(visit_date) IN (6,7,8,9) THEN 1 ELSE 0 END)
    SUM(CASE WHEN MONTH(visit_date) IN (10,11) THEN 1 ELSE 0 END) A
    COUNT(*) AS total
FROM visits
GROUP BY ayurvedic_diagnosis
ORDER BY total DESC
LIMIT 10
"""
df14 = pd.read_sql(query, conn)

fig, ax = plt.subplots(figsize=(14, 8))
df_heat = df14.set_index('ayurvedic_diagnosis')[['winter', 'spring'
sns.heatmap(df_heat, annot=True, fmt='d', cmap='YlOrRd', linewidths
ax.set_title('Adarsha Clinic — Seasonal Diagnosis Patterns', fontsi
ax.set_ylabel('')
plt.tight_layout()
plt.show()
```

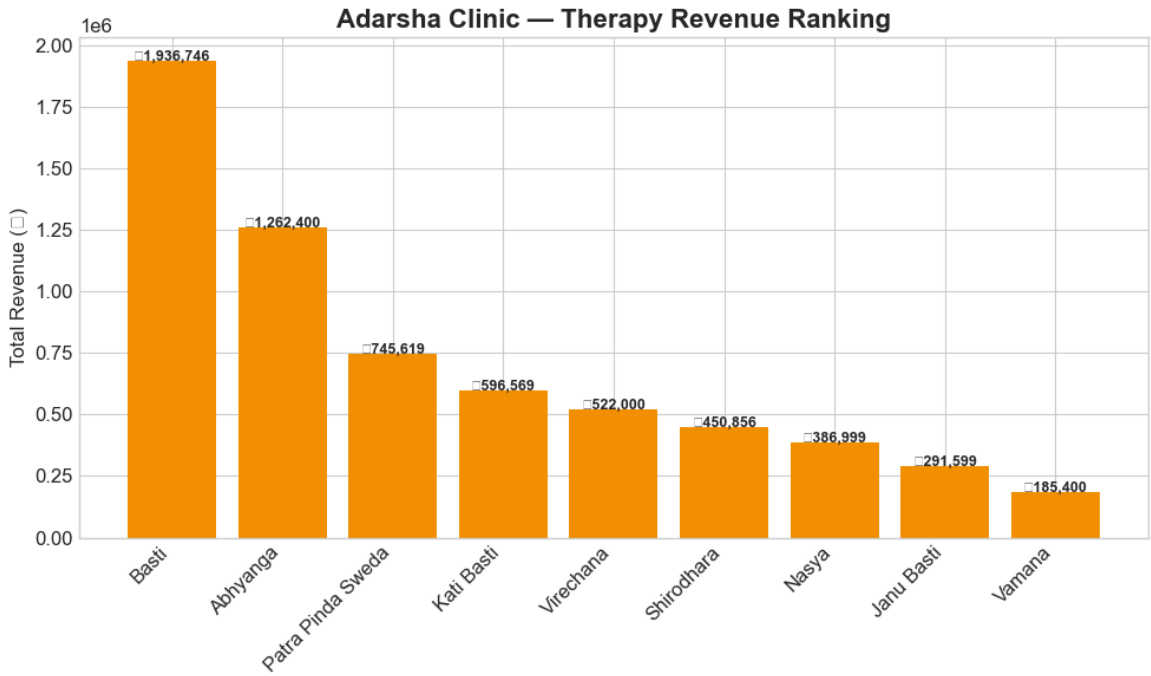
```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[20], line 18
     16 fig, ax = plt.subplots(figsize=(14, 8))
     17 df_heat = df14.set_index('ayurvedic_diagnosis')[['winter', 'spring', 'monsoon', 'post_monsoon']]
---> 18 sns.heatmap(df_heat, annot=True, fmt=   , cmap=        , linewidths=0.5, ax=ax)
     19 ax.set_title('Adarsha Clinic - Seasonal Diagnosis Patterns', fontsize=16, fontweight='bold')
     20 ax.set_ylabel('')

File /opt/anaconda3/lib/python3.13/site-packages/seaborn/matrix.py:459, in heatmap(data, vmin, vmax, cmap, center, robust, annot, fmt, annot_kws, linewidths, linecolor, cbar, cbar_kws, cbar_ax, square, xticklabels, yticklabels, mask, ax, **kwargs)
    457 if square:
    458     ax.set_aspect("equal")
--> 459 plotter.plot(ax, cbar_ax, kwargs)
    460 return ax

File /opt/anaconda3/lib/python3.13/site-packages/seaborn/matrix.py:352, in _HeatMapper.plot(self, ax, cax, kws)
    350 # Annotate the cells with the formatted values
    351 if self.annot:
--> 352     self._annotate_heatmap(ax, mesh)

File /opt/anaconda3/lib/python3.13/site-packages/seaborn/matrix.py:260, in _HeatMapper._annotate_heatmap(self, ax, mesh)
    258 lum = relative_luminance(color)
    259 text_color = ".15" if lum > .408 else "w"
--> 260 annotation = (     + self.fmt +    ).format(val)
    261 text_kwargs = dict(color=text_color, ha="center", va="center")
    262 text_kwargs.update(self.annot_kws)

ValueError: Unknown format code 'd' for object of type 'float'
```



In [21]:
```
query = """
SELECT
    DATE_FORMAT(bill_date, '%Y-%m') AS month,
    SUM(total_amount) AS monthly_revenue,
    SUM(SUM(total_amount)) OVER (ORDER BY DATE_FORMAT(bill_date, '%
FROM billing
GROUP BY month
ORDER BY month
"""
df15 = pd.read_sql(query, conn)

fig, ax1 = plt.subplots(figsize=(14, 6))
ax2 = ax1.twinx()

ax1.bar(range(len(df15)), df15['monthly_revenue'], alpha=0.5, color
ax2.plot(range(len(df15)), df15['cumulative_revenue'], color='#C73E

ax1.set_xlabel('Month')
ax1.set_ylabel('Monthly Revenue (₹)', color='#2E86AB')
ax2.set_ylabel('Cumulative Revenue (₹)', color='#C73E1D')
ax1.set_xticks(range(len(df15)))
ax1.set_xticklabels(df15['month'], rotation=45, ha='right')

plt.title('Adarsha Clinic — Cumulative Revenue Growth', fontsize=16
fig.legend(loc='upper left', bbox_to_anchor=(0.12, 0.88))
plt.tight_layout()
plt.show()
```



In [22]:
```
query = """
SELECT
    therapy_name,
    SUM(session_fee) AS total_revenue,
    ROUND(SUM(session_fee) * 100.0 / (SELECT SUM(session_fee) FROM
    RANK() OVER (ORDER BY SUM(session_fee) DESC) AS revenue_rank
FROM panchakarma_sessions
GROUP BY therapy_name
ORDER BY revenue_rank
"""
df16 = pd.read_sql(query, conn)

plt.figure(figsize=(10, 6))
plt.bar(df16['therapy_name'], df16['total_revenue'], color='#F18F01
plt.title('Adarsha Clinic — Therapy Revenue Ranking', fontsize=16,
plt.ylabel('Total Revenue (₹)')
plt.xticks(rotation=45, ha='right')
for i, v in enumerate(df16['total_revenue']):
    plt.text(i, v + 1000, f'₹{v:,.0f}', ha='center', fontsize=9, fo
plt.tight_layout()
plt.show()

print(df16.to_string(index=False))
```



therapy_name	total_revenue	revenue_share_pct	revenue_rank
Basti	1936746.36	30.4	1
Abhyanga	1262400.00	19.8	2
Patra Pinda Sweda	745618.71	11.7	3
Kati Basti	596569.38	9.4	4
Virechana	522000.00	8.2	5
Shirodhara	450856.36	7.1	6
Nasya	386999.31	6.1	7
Janu Basti	291598.92	4.6	8
Vamana	185400.00	2.9	9

In [23]:

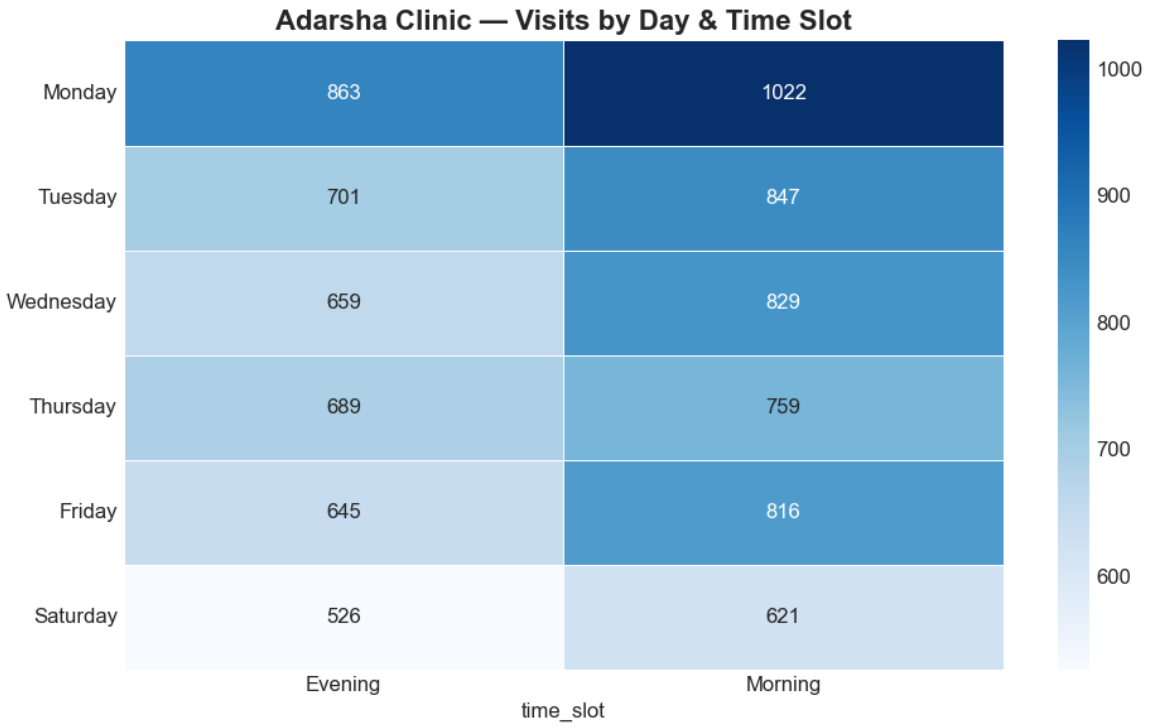
```
query = """
WITH visit_gaps AS (
    SELECT
        patient_id,
        visit_date,
        LAG(visit_date) OVER (PARTITION BY patient_id ORDER BY visi
        DATEDIFF(visit_date, LAG(visit_date) OVER (PARTITION BY pat
    FROM visits
)
SELECT
    vg.patient_id,
    CONCAT(p.first_name, ' ', p.last_name) AS name,
    COUNT(*) AS total_visits,
    ROUND(AVG(gap_days), 1) AS avg_days_between_visits
FROM visit_gaps vg
JOIN patients p ON vg.patient_id = p.patient_id
WHERE gap_days IS NOT NULL
GROUP BY vg.patient_id, name
HAVING total_visits >= 5
ORDER BY avg_days_between_visits ASC
LIMIT 20
"""
df17 = pd.read_sql(query, conn)

print("📊 Adarsha Clinic — Most Frequent Returning Patients")
print("=" * 65)
```

```
print(df17.to_string(index=False))
```

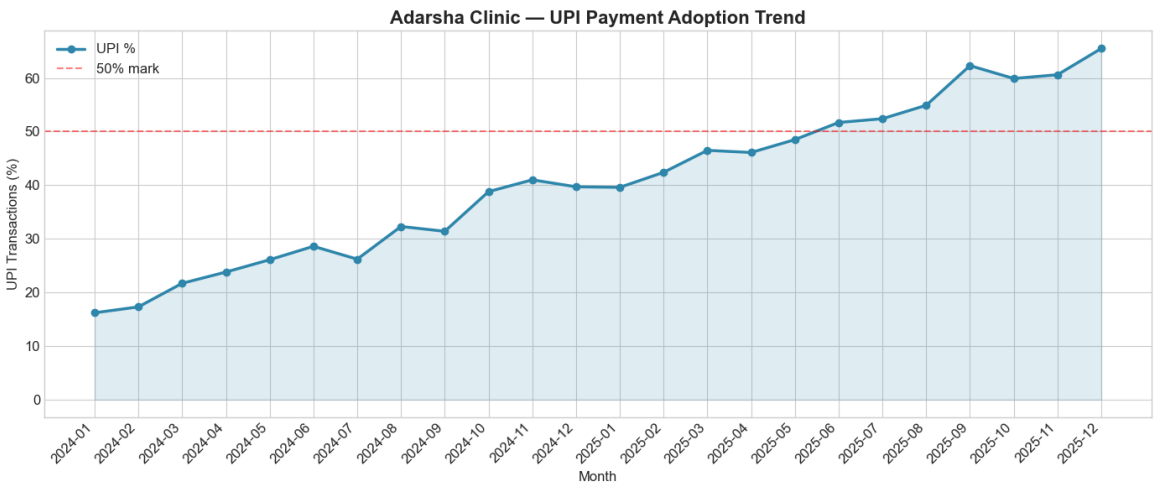
```
📊 Adarsha Clinic — Most Frequent Returning Patients
====================================================================
 patient_id             name  total_visits  avg_days_between_visits
       2218     Ratna Murthy             9                     10.9
       3356   Gangamma Gupta             8                     14.0
       3714   Vinod Hiremath            11                     17.1
       2048       Umesh Bhat             5                     18.0
       1470   Bharathi Hugar             9                     22.6
       3411      Umesh Gupta             5                     23.6
       2721   Bharathi Mohan             9                     24.0
       3212       Asha Gupta             9                     24.1
       2412  Basavaraj Hegde             7                     24.3
       3517 Suresh Doddamani             6                     24.3
       1533  Kamalamma Reddy             5                     26.0
       2346     Eramma Hegde            14                     26.1
       2519  Dinesh Havaldar             7                     26.7
       2229  Kavitha Hadapad             9                     28.4
       2141  Manjunath Gupta             7                     29.1
       3845    Devappa Verma            13                     29.2
       1003   Rukmini Gangur            12                     30.3
       3711    Priya Hadapad             6                     31.0
       1417      Roopa Gupta            11                     31.5
       3507     Suresh Joshi             6                     31.7
```

In [24]:

```
query = """
SELECT
    DAYNAME(visit_date) AS day_of_week,
    time_slot,
    COUNT(*) AS visits
FROM visits
GROUP BY day_of_week, time_slot
ORDER BY FIELD(day_of_week, 'Monday','Tuesday','Wednesday','Thursda
"""
df18 = pd.read_sql(query, conn)

df_pivot = df18.pivot(index='day_of_week', columns='time_slot', val
day_order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday'
df_pivot = df_pivot.reindex(day_order)

fig, ax = plt.subplots(figsize=(10, 6))
sns.heatmap(df_pivot, annot=True, fmt='d', cmap='Blues', linewidths
ax.set_title('Adarsha Clinic — Visits by Day & Time Slot', fontsize
ax.set_ylabel('')
plt.tight_layout()
plt.show()
```



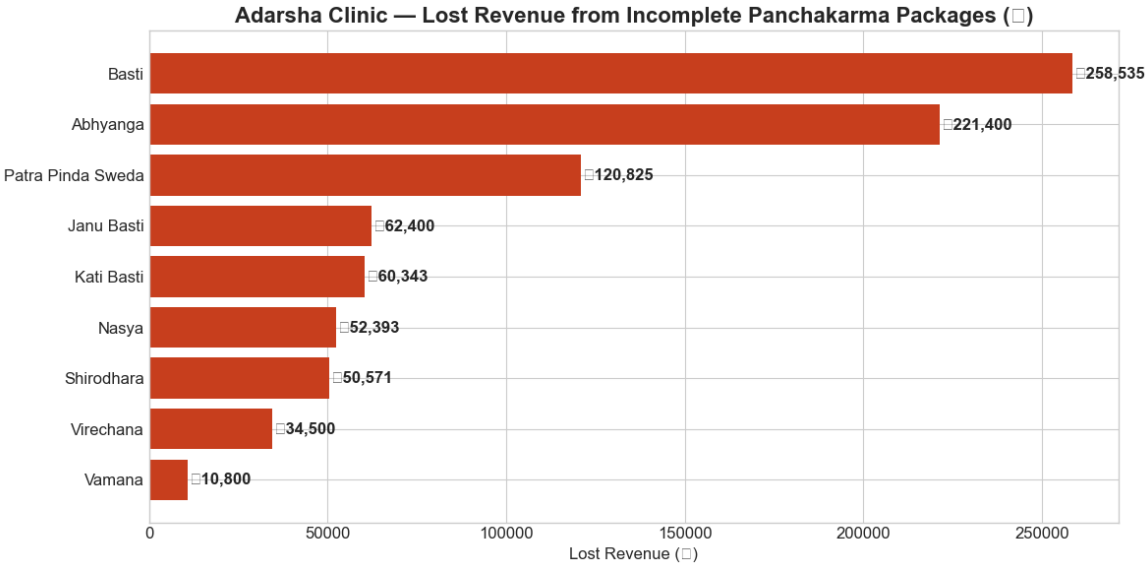
In [25]:
```
query = """
SELECT
    DATE_FORMAT(bill_date, '%Y-%m') AS month,
    COUNT(*) AS total_bills,
    SUM(CASE WHEN payment_method = 'UPI' THEN 1 ELSE 0 END) AS upi_
    SUM(CASE WHEN payment_method = 'Cash' THEN 1 ELSE 0 END) AS cas
    ROUND(SUM(CASE WHEN payment_method = 'UPI' THEN 1 ELSE 0 END) *
FROM billing
GROUP BY month
ORDER BY month
"""
df19 = pd.read_sql(query, conn)

plt.figure(figsize=(14, 6))
plt.plot(df19['month'], df19['upi_pct'], marker='o', linewidth=2.5,
plt.fill_between(range(len(df19)), df19['upi_pct'], alpha=0.15, col
plt.title('Adarsha Clinic — UPI Payment Adoption Trend', fontsize=1
plt.xlabel('Month')
plt.ylabel('UPI Transactions (%)')
plt.xticks(range(len(df19)), df19['month'], rotation=45, ha='right'
plt.axhline(y=50, color='red', linestyle='--', alpha=0.5, label='50
plt.legend()
plt.tight_layout()
plt.show()

print(f"UPI % in Jan 2024: {df19.iloc[0]['upi_pct']}%")
print(f"UPI % in Dec 2025: {df19.iloc[-1]['upi_pct']}%")
```




```
UPI % in Jan 2024: 16.2%
UPI % in Dec 2025: 65.5%
```

In [26]:

```
query = """
WITH package_status AS (
    SELECT
        patient_id,
        visit_id,
        therapy_name,
        total_sessions,
        SUM(CASE WHEN status = 'Completed' THEN 1 ELSE 0 END) AS co
        SUM(CASE WHEN status IN ('Missed', 'Cancelled') THEN 1 ELSE
        SUM(CASE WHEN status IN ('Missed', 'Cancelled') THEN sessio
    FROM panchakarma_sessions
    GROUP BY patient_id, visit_id, therapy_name, total_sessions
    HAVING missed > 0
)
SELECT
    therapy_name,
    COUNT(*) AS incomplete_packages,
    SUM(missed) AS total_missed_sessions,
    SUM(lost_revenue) AS total_lost_revenue,
    ROUND(AVG(completed * 100.0 / total_sessions), 1) AS avg_comple
FROM package_status
GROUP BY therapy_name
ORDER BY total_lost_revenue DESC
"""
df20 = pd.read_sql(query, conn)

plt.figure(figsize=(12, 6))
plt.barh(df20['therapy_name'][::-1], df20['total_lost_revenue'][::-
plt.title('Adarsha Clinic — Lost Revenue from Incomplete Panchakarm
plt.xlabel('Lost Revenue (₹)')
for i, v in enumerate(df20['total_lost_revenue'][::-1]):
    plt.text(v + 500, i, f'₹{v:,.0f}', va='center', fontweight='bol
plt.tight_layout()
plt.show()

total_lost = df20['total_lost_revenue'].sum()
print(f"\n⚠️ Total Lost Revenue from Incomplete Packages: ₹{total_l
print(f"\n{df20.to_string(index=False)}")
```




```
⚠️ Total Lost Revenue from Incomplete Packages: ₹871,767

     therapy_name  incomplete_packages  total_missed_sessions  total
_lost_revenue  avg_completion_pct
            Basti                   83                  318.0
258535.18                56.5
         Abhyanga                   94                  369.0
221400.00                54.9
Patra Pinda Sweda                   39                  164.0
120825.15                50.7
       Janu Basti                   23                   96.0
62399.76                52.8
       Kati Basti                   31                   90.0
60342.54                62.2
            Nasya                   34                  128.0
52392.74                55.9
       Shirodhara                   22                   61.0
50571.30                65.6
        Virechana                   23                   23.0
34500.00                 0.0
           Vamana                    6                    6.0
10800.00                 0.0
```

In [27]:

```
# Gather all key numbers
total_revenue = pd.read_sql("SELECT SUM(total_amount) AS val FROM b
total_patients = pd.read_sql("SELECT COUNT(*) AS val FROM patients"
total_visits = pd.read_sql("SELECT COUNT(*) AS val FROM visits", co
total_sessions = pd.read_sql("SELECT COUNT(*) AS val FROM panchakar
completion_rate = pd.read_sql("SELECT ROUND(SUM(CASE WHEN status='C
lost_revenue = pd.read_sql("SELECT SUM(session_fee) AS val FROM pan

rev_by_year = pd.read_sql("SELECT YEAR(bill_date) AS yr, SUM(total_
yoy_growth = round((rev_by_year.iloc[1]['rev'] - rev_by_year.iloc[0

print("=" * 60)
print("   ADARSHA CLINIC — EXECUTIVE SUMMARY")
print("   Ayurvedic Clinic Operations Dashboard")
print("   Period: January 2024 — December 2025")
print("=" * 60)
print(f"\n   👥 Total Patients:          {total_patients:,}")
```

```
print(f"   📋 Total Visits:            {total_visits:,}")
print(f"   🧘 Panchakarma Sessions:    {total_sessions:,}")
print(f"   💰 Total Revenue:           ₹{total_revenue:,.0f}")
print(f"   📈 YoY Revenue Growth:      {yoy_growth}%")
print(f"   ✅ Panchakarma Completion:   {completion_rate}%")
print(f"   ⚠️  Lost Revenue (Dropouts): ₹{lost_revenue:,.0f}")
print(f"\n   Revenue 2024: ₹{rev_by_year.iloc[0]['rev']:,.0f}")
print(f"   Revenue 2025: ₹{rev_by_year.iloc[1]['rev']:,.0f}")
print("=" * 60)
```

```
============================================================
   ADARSHA CLINIC — EXECUTIVE SUMMARY
   Ayurvedic Clinic Operations Dashboard
   Period: January 2024 — December 2025
============================================================

   👥 Total Patients:          3,000
   📋 Total Visits:            8,977
   🧘 Panchakarma Sessions:    8,789
   💰 Total Revenue:           ₹8,311,647
   📈 YoY Revenue Growth:      19.2%
   ✅ Panchakarma Completion:   85.7%
   ⚠️  Lost Revenue (Dropouts): ₹871,767

   Revenue 2024: ₹3,792,038
   Revenue 2025: ₹4,519,609
============================================================
```

In [29]:

```
conn.close()
print("✅ MySQL connection closed")
print("\n🎉 Adarsha Clinic — Operations Analysis Complete!")
print("📁 Save this notebook and export as PDF for your report")
```

```
✅ MySQL connection closed

🎉 Adarsha Clinic — Operations Analysis Complete!
📁 Save this notebook and export as PDF for your report
```

In [ ]: